

LILLEHAMMER-SAETHERENGEN, Norway

WMO: 013780

Lat: 61.093N

Lon: 10.477E

Elev: 784

StdP: 14.28

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	1.5	-7.0	3.9	-1.7	-3.2	4.8	2.1	17.0	34.0	13.8	35.8	3.7	10	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	79.4	61.4	75.6	60.2	71.8	58.7	63.9	73.5	62.2	70.7	60.6	68.8	5.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	81.4	66.1	58.9	76.5	65.1	57.2	71.8	63.7	29.5	73.6	28.3	70.6	27.1	68.7	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.0	10.0	-5.1	83.6	5.4	3.6	-9.0	86.1	-12.2	88.2	-15.2	90.2	-19.1	92.8		
			-5.6	66.5	5.3	2.2	-9.4	68.1	-12.5	69.4	-15.5	70.6	-19.4	72.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	19.6	22.3	30.6	40.2	49.5	57.3	61.8	59.0	51.3	39.6	30.1	22.3	
	DBStd	16.40	9.79	9.20	8.09	5.77	7.13	5.42	5.43	4.54	5.39	6.59	7.69	10.21	
	HDD50	4553	942	775	601	297	97	5	0	1	49	329	597	860	
	HDD65	9033	1407	1195	1066	743	483	240	134	193	411	789	1047	1325	
	CDD50	1047	0	0	0	4	81	224	366	279	88	5	0	0	
	CDD65	52	0	0	0	0	2	9	35	6	0	0	0	0	
	CDH74	469	0	0	0	0	37	110	279	43	0	0	0	0	
	CDH80	60	0	0	0	0	1	11	47	1	0	0	0	0	
Wind	WSAvg	4.2	3.8	3.9	4.3	4.5	4.9	4.8	4.2	3.9	4.1	4.0	4.0	4.2	
Precipitation	PrecAvg	32.5	2.4	1.7	1.7	1.8	3.0	3.1	4.0	4.1	2.7	3.2	3.5	2.3	
	PrecMax	38.5	5.0	4.2	3.8	3.1	5.9	6.2	8.5	6.3	5.1	9.7	12.3	3.8	
	PrecMin	23.0	0.7	0.5	0.0	0.6	0.2	0.3	0.6	1.0	1.2	0.6	0.8	0.9	
	PrecStd	4.1	1.1	0.9	1.0	0.8	1.5	1.6	1.7	1.6	1.1	2.1	2.4	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.9	44.3	53.3	64.6	78.9	81.6	84.1	78.1	69.2	60.2	51.5	44.9	
		MCWB	38.2	38.1	42.9	49.5	58.1	60.8	63.8	62.3	57.1	53.1	48.1	40.4	
	2%	DB	38.2	39.7	48.3	59.5	72.2	77.5	80.9	74.6	65.7	54.9	46.6	41.9	
		MCWB	34.8	35.9	39.6	47.0	55.5	59.2	63.1	61.4	55.8	49.6	43.3	38.4	
	5%	DB	35.5	36.5	44.5	54.9	67.5	72.4	77.2	71.1	62.7	52.0	43.6	39.1	
		MCWB	33.2	33.8	37.1	44.3	53.4	57.5	61.7	59.6	54.6	47.5	41.1	35.8	
	10%	DB	33.1	34.1	41.0	50.8	62.2	68.2	72.8	67.8	60.0	49.3	40.1	36.1	
		MCWB	31.5	31.8	35.4	41.9	50.9	55.0	60.2	58.2	53.4	45.6	38.0	33.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	39.4	43.4	51.2	60.3	62.9	67.1	65.0	60.1	54.3	48.4	41.4
			MCDB	41.4	43.0	51.9	61.3	72.3	71.7	78.7	73.2	65.1	58.3	51.8	44.1
2%		WB	35.6	36.1	40.3	47.8	57.6	61.1	64.9	63.0	57.9	50.8	44.2	38.7	
		MCDB	37.5	38.8	46.8	56.9	69.0	72.4	75.8	70.3	63.0	54.0	46.0	41.4	
5%		WB	33.5	34.2	37.9	45.4	54.7	59.2	63.1	61.4	56.1	48.3	41.3	36.0	
		MCDB	35.2	36.5	43.2	53.6	64.1	69.7	73.0	68.0	60.8	51.0	43.2	38.5	
10%		WB	31.5	32.1	35.9	43.1	52.2	57.2	61.5	59.7	54.4	46.2	38.1	33.8	
		MCDB	33.1	33.7	40.4	49.6	60.3	66.7	70.3	65.7	58.7	49.0	39.9	35.9	
Mean Daily Temperature Range		5% DB	MDBR	8.8	10.0	14.0	15.9	16.8	17.2	16.4	15.6	14.5	10.1	7.7	8.4
			MCDBR	10.7	12.9	18.1	23.2	24.9	24.1	22.6	20.6	18.2	13.6	12.1	12.6
	MCWBR		8.9	9.5	11.2	13.0	12.0	11.6	9.9	10.1	10.4	9.6	9.7	10.5	
	5% WB	MCDBR	10.5	11.5	16.8	21.5	20.7	20.4	19.1	16.6	15.5	11.3	11.0	11.5	
		MCWBR	9.2	8.7	10.8	12.5	10.7	10.7	9.1	9.1	9.8	8.9	9.5	9.9	
Clear-Sky Solar Irradiance	taub		0.225	0.272	0.304	0.344	0.350	0.344	0.361	0.353	0.319	0.303	0.242	0.191	
	taud		2.143	2.254	2.252	2.274	2.416	2.472	2.437	2.454	2.532	2.484	2.226	2.026	
	Ebn at Noon		168	228	259	269	277	281	272	265	256	219	159	124	
	Edh at Noon		11	20	29	34	32	31	31	29	22	16	10	8	
All-Sky Solar Radiation	RadAvg		89	298	786	1296	1507	1682	1522	1183	771	344	111	52	
	RadStd		15	58	75	138	149	152	194	105	93	62	27	5	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page